

Qatar-1b

R-2584

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260715 · created 2026-08-27T01:17:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461236.8529 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.056
Transit depth (Rp/Rs)^2	3.7 +/- 2.15 [%]
Orbital Inclination (inc)	88.51 +/- 2.83
Airmass coefficient 1 (a1)	3.03 +/- 0.34
Airmass coefficient 2 (a2)	-0.85 +/- 1.09
Scatter in the residuals of the lightcurve fit is	11.85 %
Best Comparison Star	#3 - [209, 257]
Optimal Aperture	4.66
Optimal Annulus	13.06
Transit Duration (day)	0.0826 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.056 vs 0.1463 (deviation 0.53σ)
Duration fit vs published	0.0826 d vs 0.0692 d (deviation 1.21σ)
Mid-time O–C	5.3 min
Depth SNR	2.2
Residual scatter	11.85 %

Light curve

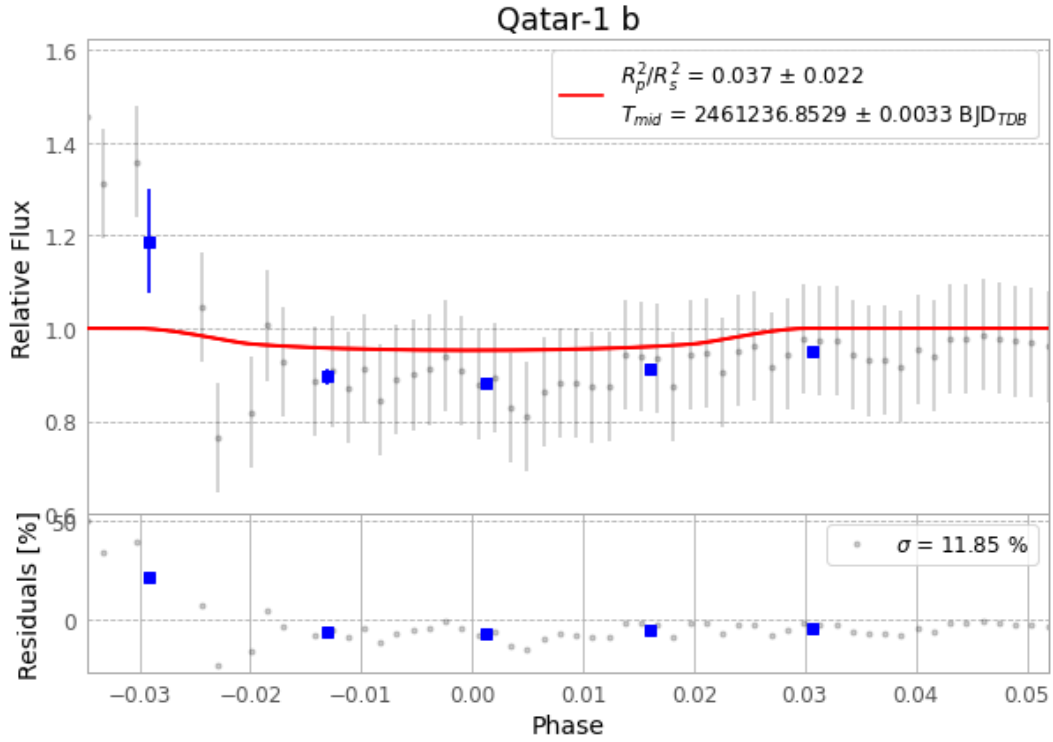


Figure 1 — FinalLightCurve_Qatar-1 b_2026-07-15.png (SHA-256 e69fccddf935bae1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d30e67dd25b25fe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), Qatar-1260715071515.FITS → Qatar-1260715101215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260715.log (159575240b9e...), FinalLightCurve_Qatar-1 b_2026-07-15.png (e69fccddf935...), FinalLightCurve_Qatar-1 b_2026-07-15.csv (fe2e8fd71a78...)

Compute resources

Wall time 210.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 01:17Z.